

Final Presentation: Group 1

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Research Question

- How do squirrel-human interactions differ across various NYC parks?
 - Interested in environmental and atmosphere/ambiance factors
 - Squirrels are important part of the NYC ecosystem & health of urban green spaces
- Our hypothesis: Busier parks will have braver squirrels!
 - Adaptation from exposure

Our Dataset

- Dataset source: The Squirrel Census sample from March 1st, 2020
- 3 Datasets: Park dataset, Squirrel dataset, & Park Info dataset (NYCgovparks.org, made by us)
 - Left-joined by common variables (area name, area ID, park name, and park ID)
- Cleaned data mainly using grepl() function
 - for some variables, people were allowed to write additional comments

Our Dataset (cont.)

- Park Dataset:
 - Park conditions (boolean: calm or busy)
 - Area name: area where the park is located (upper manhattan, central manhattan, lower manhattan, or brooklyn)
 - Area ID: A, B, C, or D assigned to an area name
 - Park name: name of the park
 - Park ID: 1-24 assigned to park names
 - Number of squirrels: squirrel count at specific park
 - Litter (some, none, abundant)

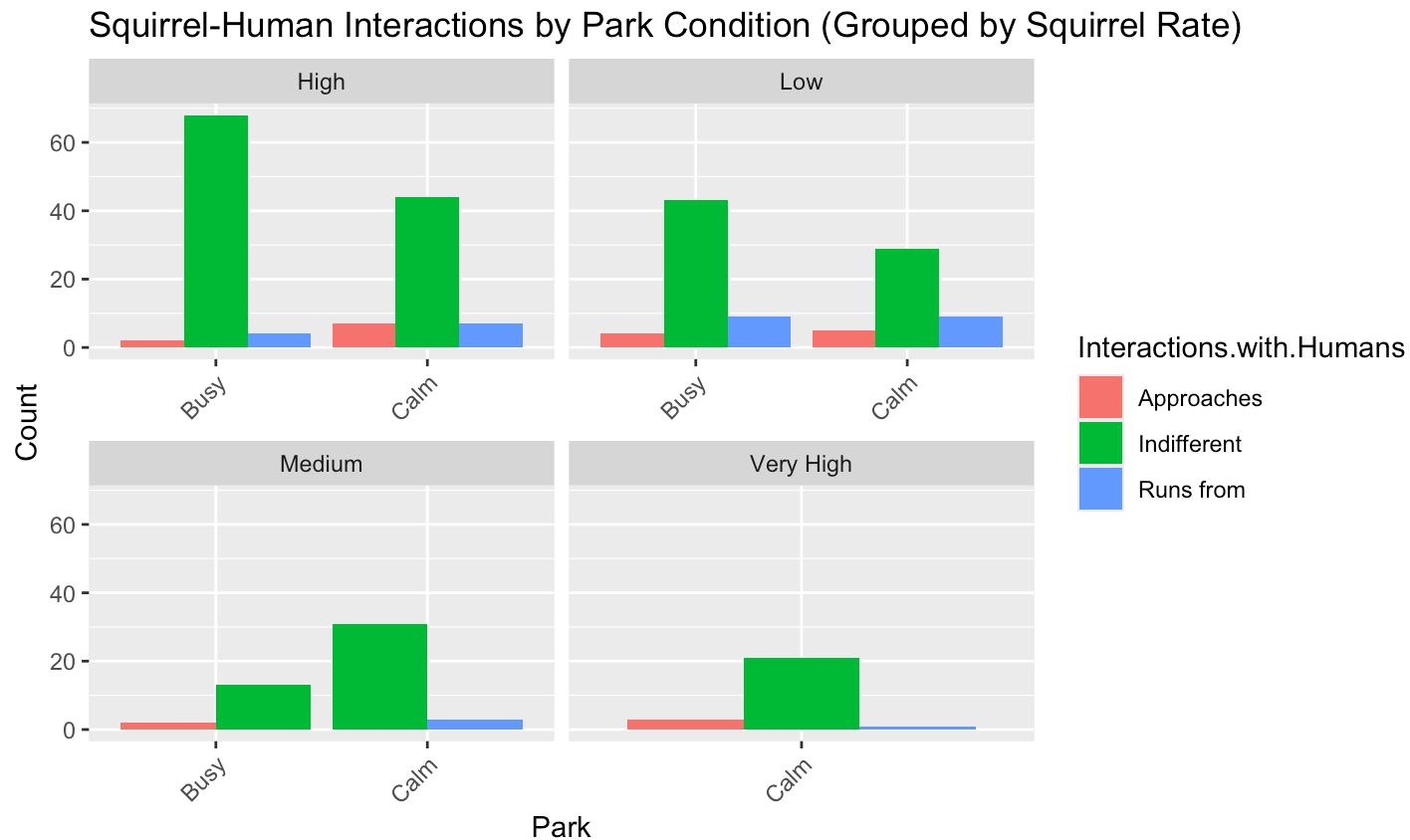
Our Dataset (cont.)

- Squirrel Dataset:
 - Area name, Area ID, Park name, Park ID
 - Primary fur color: gray, black, or cinnamon
 - Squirrel longitude: longitude coordinate for observed squirrel
 - Squirrel latitude: latitude coordinate for observed squirrel
 - Interactions with Humans: approaches, indifferent, or runs from
 - Activities: value is either “running”, “chasing”, “climbing”, or “foraging”
 - Location: location when first sighted (value is either “ground plane”, “above ground”, or specific location)

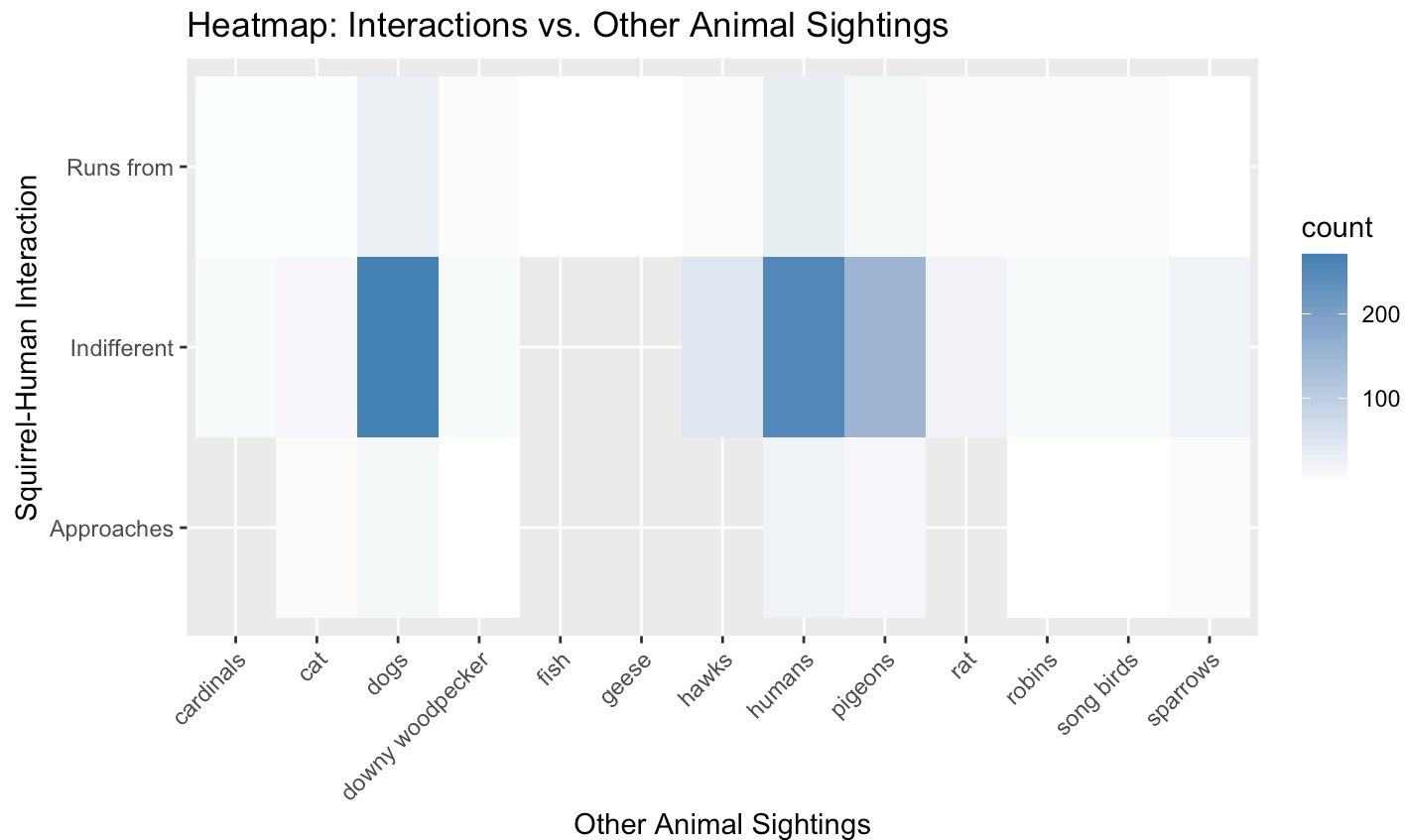
Our Dataset (cont.)

- Park Info Dataset:
 - Park name
 - Park ID
 - Total acres
 - Acres of natural areas
 - Number of mapped trees
- Helped us group parks
- We manually created the following variables:
 - Squirrel Rate: $\text{Number of Squirrels} / \text{Total Acres}$
 - Rate Group: Quantiles of Squirrel Rate
 - Park Size: Quantiles of Total Acres

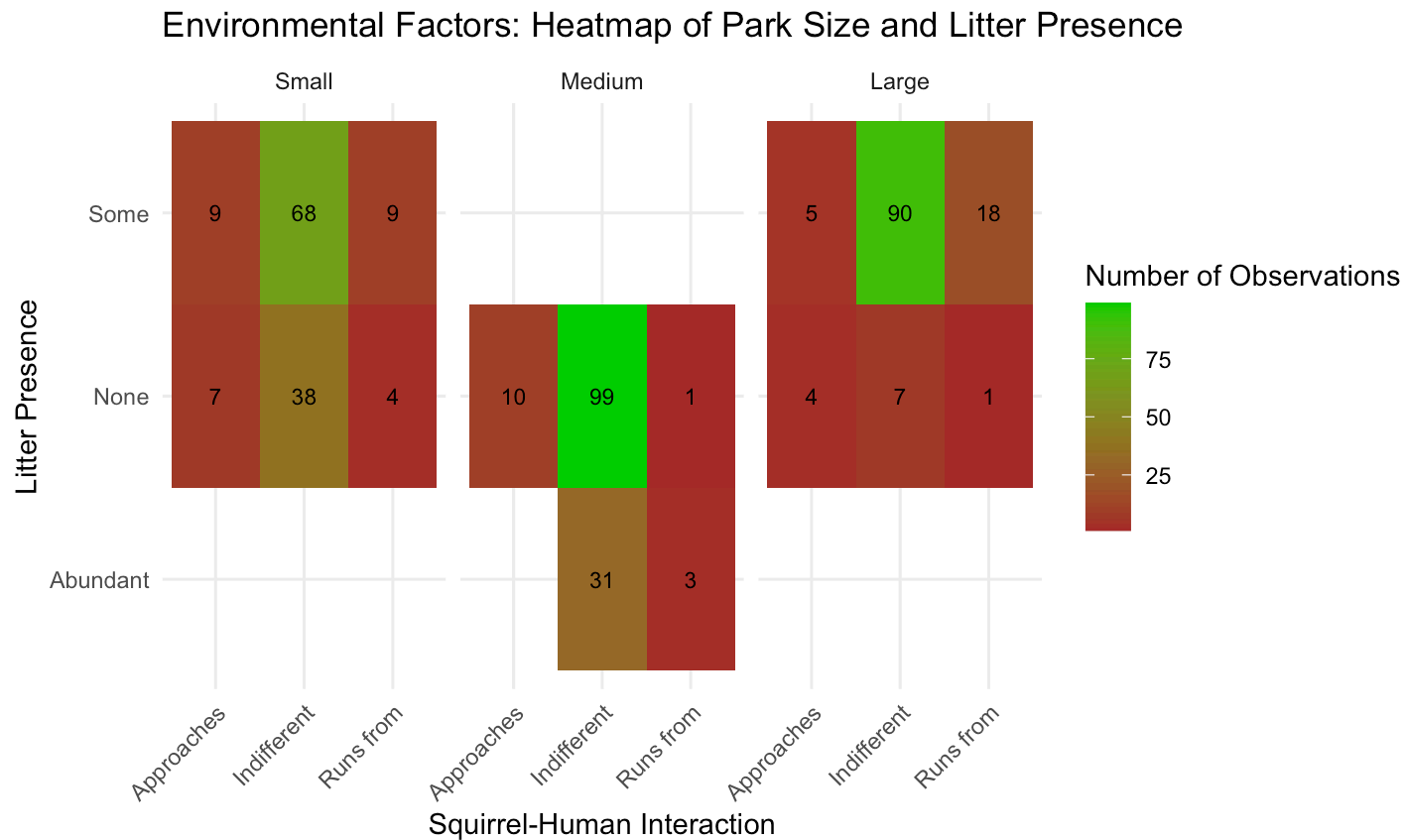
Atmosphere/ambiance: Stacked Barplot



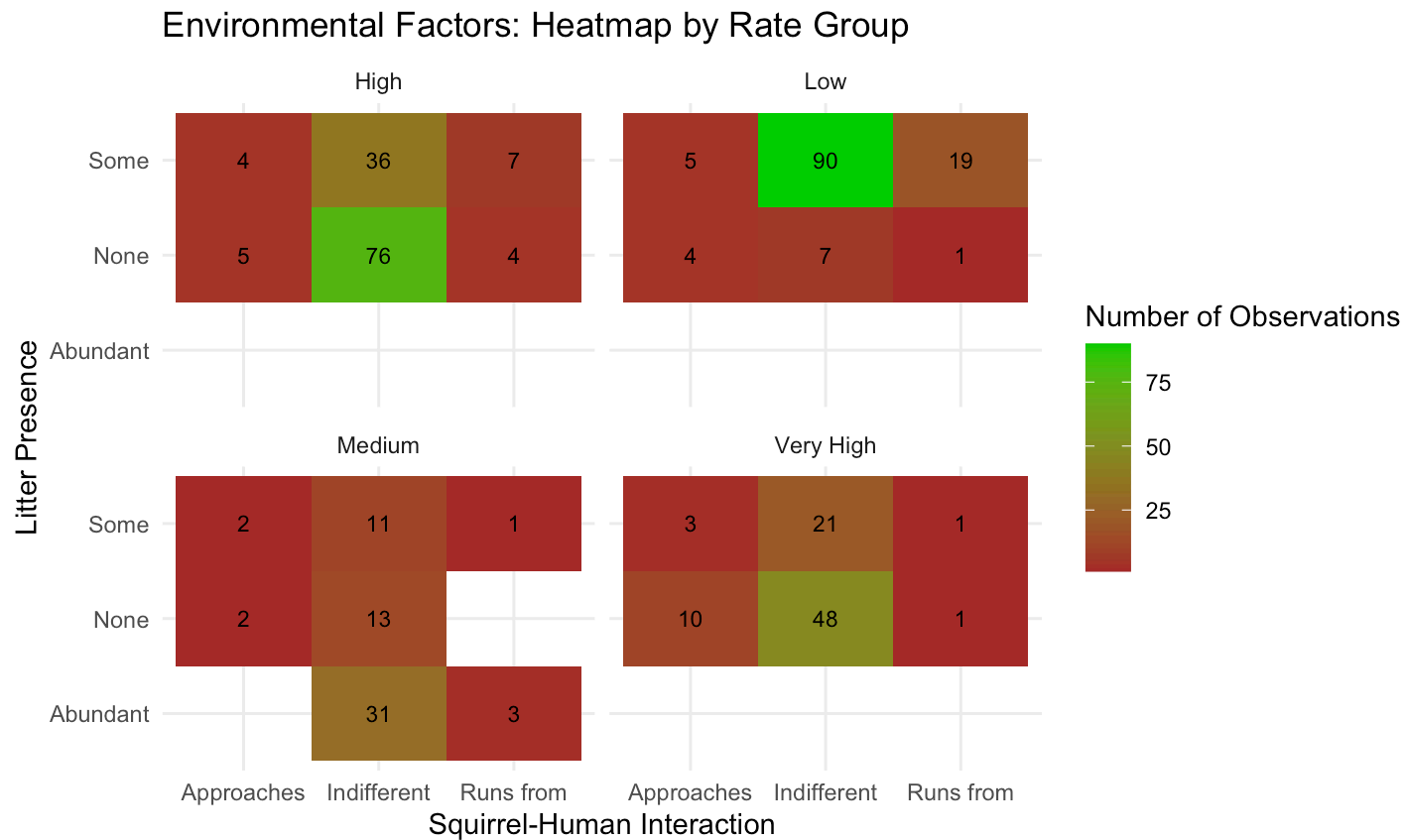
Atmosphere/ambiance: Heatmap



Environment: Heatmap



Environment: Heatmap (cont.)



Squirrel Map: Shiny App

- Parks in Lower Manhattan tend to be busier than parks in Upper Manhattan.
 - We also observe that squirrels in Lower Manhattan are more likely to approach humans than those in Upper Manhattan.
- Many had missing data for longitude and latitude, so we estimated using different methods.
- https://admedeiros.shinyapps.io/Final_Project/